

Quasi-Monte Carlo integration for Bayesian design of experiment problems governed by PDEs

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Table of contents

Part I: Quasi-Monte Carlo methods

Part II: Bayesian optimal experimental design

Part I: Quasi-Monte Carlo methods

High-dimensional numerical integration

$$\int_{[0,1]^s} f(\mathbf{y}) d\mathbf{y} \approx \sum_{i=1}^n w_i f(\mathbf{t}_i)$$

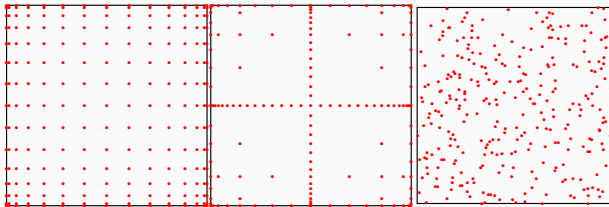


Figure 1: Tensor product grid, sparse grid, Monte Carlo nodes (not QMC rules)

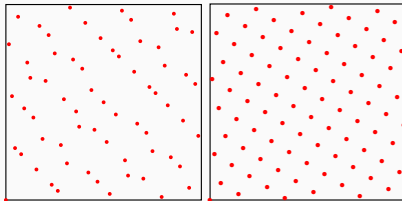


Figure 2: Sobol' points, lattice rule (examples of QMC rules)

Quasi-Monte Carlo (QMC) methods are a class of *equal weight* cubature rules

$$\int_{[0,1]^s} f(\mathbf{y}) \, d\mathbf{y} \approx \frac{1}{n} \sum_{i=1}^n f(\mathbf{t}_i),$$

where $(\mathbf{t}_i)_{i=1}^n$ is an ensemble of *deterministic* nodes in $[0, 1]^s$.

The nodes $(\mathbf{t}_i)_{i=1}^n$ are NOT random!! Instead, they are *deterministically chosen*.

QMC methods exploit the smoothness and anisotropy of an integrand in order to achieve better-than-Monte Carlo rates.

Lattice rules

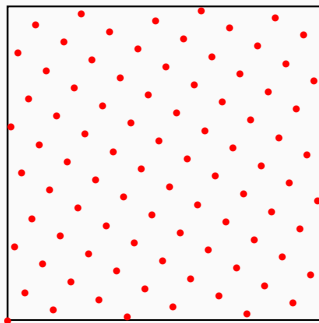
Rank-1 lattice rules

$$Q_{s,n}(f) = \frac{1}{n} \sum_{i=1}^n f(\mathbf{t}_i)$$

have the points

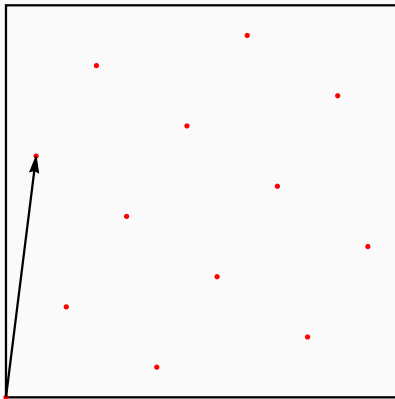
$$\mathbf{t}_i = \text{mod} \left(\frac{i\mathbf{z}}{n}, 1 \right), \quad i \in \{1, \dots, n\},$$

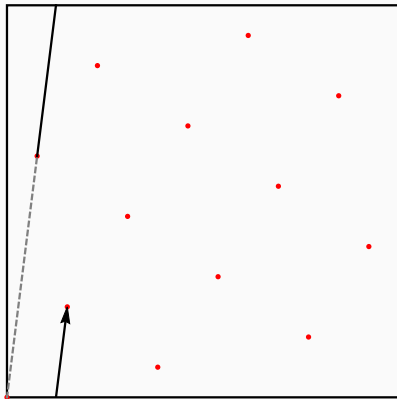
where the entire point set is determined by the *generating vector* $\mathbf{z} \in \mathbb{N}^s$, with all components *coprime* to n .

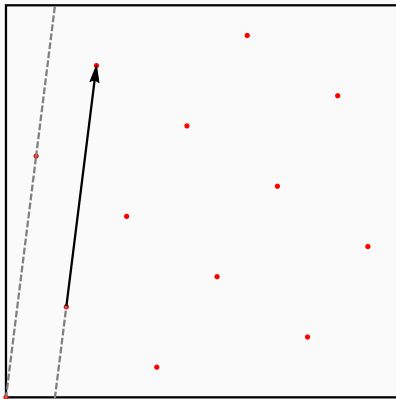


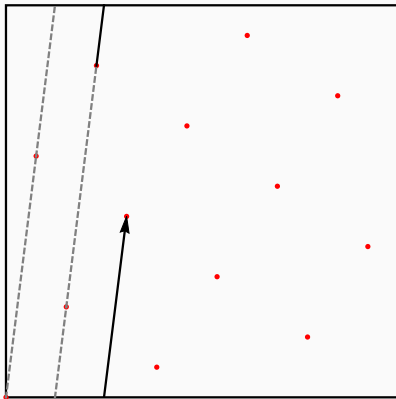
Lattice rule with $\mathbf{z} = (1, 55)$ and $n = 89$
nodes in $[0, 1]^2$

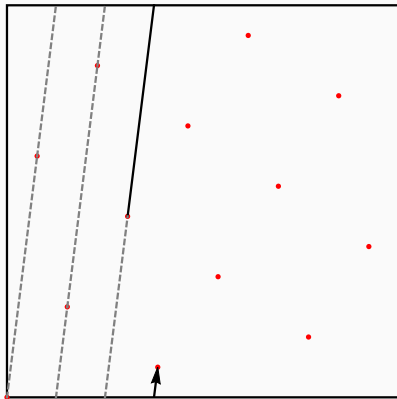
The quality of the lattice rule is determined by the choice of $\mathbf{z}$.

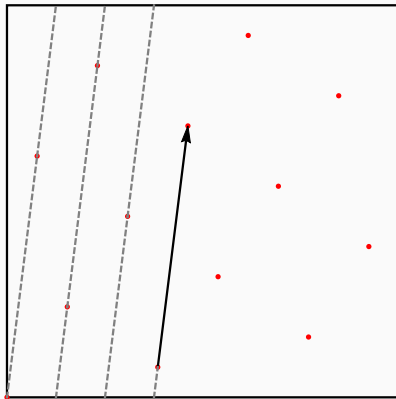


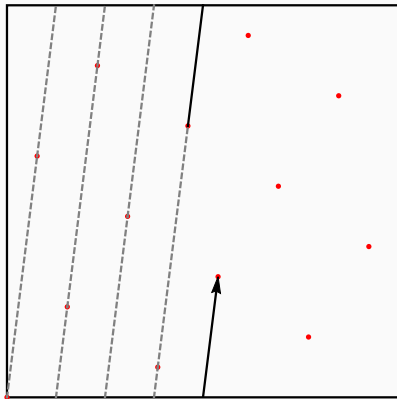


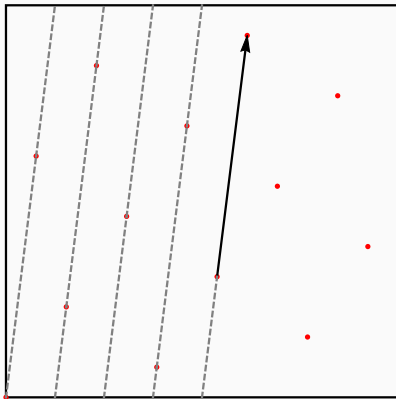


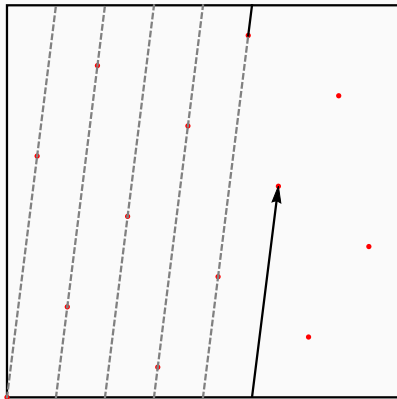


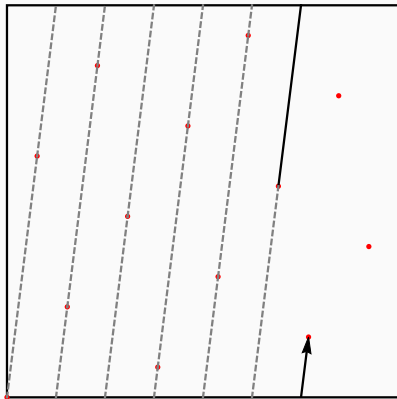


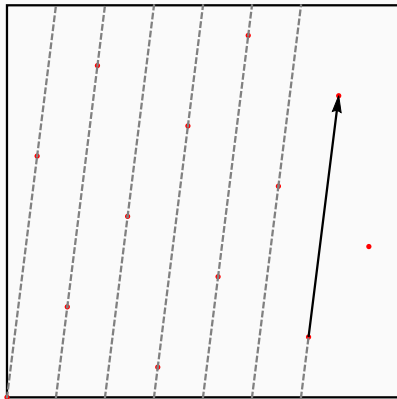


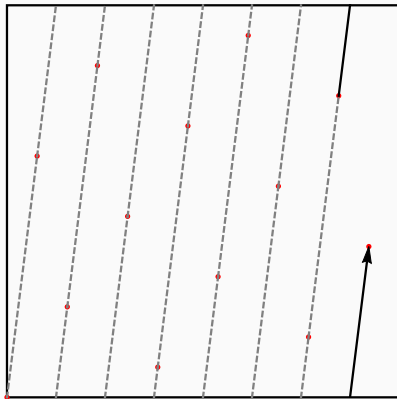












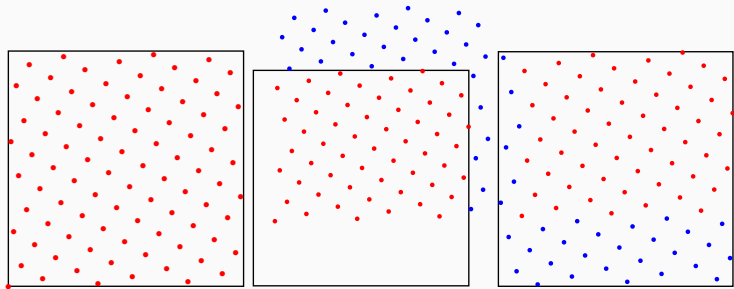
Randomly shifted lattice rules

Shifted rank-1 lattice rules have points

$$\mathbf{t}_i = \text{mod} \left(\frac{i\mathbf{z}}{n} + \mathbf{\Delta}, 1 \right), \quad i \in \{1, \dots, n\}.$$

$\mathbf{\Delta} \in [0, 1)^s$ is the *shift*

Use a number of random shifts for error estimation.



Lattice rule shifted by $\mathbf{\Delta} = (0.1, 0.3)$.

Let $\mathbf{\Delta}^{(r)}$, $r = 1, \dots, R$, be independent random shifts drawn from $U([0, 1]^s)$ and define

$$Q_{s,n}^{(r)}(f) := \frac{1}{n} \sum_{i=1}^n f(\text{mod}(\mathbf{t}_i + \mathbf{\Delta}^{(r)}, 1)). \quad (\text{QMC rule with 1 random shift})$$

Then

$$\overline{Q}_{s,n}(f) = \frac{1}{R} \sum_{r=1}^R Q_{s,n}^{(r)} f \quad (\text{QMC rule with } R \text{ random shifts})$$

is an unbiased estimator of $I_s(f)$.

Let $f: [0, 1]^s \rightarrow \mathbb{R}$ be sufficiently smooth.

Error bound (one random shift):

$$|I_s(f) - Q_{s,n}^{\Delta}(f)| \leq e_{s,n,\gamma}^{\Delta}(\mathbf{z}) \|f\|_{\gamma}.$$

R.M.S. error bound (shift-averaged):

$$\sqrt{\mathbb{E}_{\Delta}[|I_s(f) - \bar{Q}_{s,n}(f)|^2]} \leq e_{s,n,\gamma}^{\text{sh}}(\mathbf{z}) \|f\|_{\gamma}.$$

We consider weighted Sobolev spaces with dominating mixed smoothness, equipped with norm

$$\|f\|_{\gamma}^2 = \sum_{\mathbf{u} \subseteq \{1:s\}} \frac{1}{\gamma_{\mathbf{u}}} \int_{[0,1]^{|\mathbf{u}|}} \left(\int_{[0,1]^{s-|\mathbf{u}|}} \frac{\partial^{|\mathbf{u}|} f}{\partial \mathbf{y}_{\mathbf{u}}}(\mathbf{y}) d\mathbf{y}_{-\mathbf{u}} \right)^2 d\mathbf{y}_{\mathbf{u}}$$

and (squared) worst case error

$$P(\mathbf{z}) := e_{s,n,\gamma}^{\text{sh}}(\mathbf{z})^2 = \frac{1}{n} \sum_{k=0}^{n-1} \sum_{\emptyset \neq \mathbf{u} \subseteq \{1:s\}} \gamma_{\mathbf{u}} \prod_{j \in \mathbf{u}} \omega\left(\left\{\frac{kz_j}{n}\right\}\right)$$

where $\omega(x) = x^2 - x + \frac{1}{6}$.

CBC algorithm (Sloan, Kuo, Joe 2002)

The idea of the *component-by-component* (CBC) algorithm is to find a good generating vector $\mathbf{z} = (z_1, \dots, z_s)$ by proceeding as follows:

1. Set $z_1 = 1$;
2. With z_1 fixed, choose z_2 to minimize error criterion $P(z_1, z_2)$;
3. With z_1 and z_2 fixed, choose z_3 to minimize error criterion $P(z_1, z_2, z_3)$
- $\vdots$

Efficient implementation using FFT (QMC4PDE, QMCPy, etc.) if weights have certain structure (e.g., POD weights).

Theorem (CBC error bound)

The generating vector $\mathbf{z} \in \mathbb{N}^s$ constructed by the CBC algorithm, minimizing the squared shift-averaged worst-case error $[e_{s,n,\gamma}^{\text{sh}}(\mathbf{z})]^2$ for the weighted unanchored Sobolev space in each step, satisfies

$$[e_{s,n,\gamma}^{\text{sh}}(\mathbf{z})]^2 \leq \left(\frac{1}{\varphi(n)} \sum_{\emptyset \neq \mathbf{u} \subseteq \{1:s\}} \gamma_{\mathbf{u}}^{\lambda} \left(\frac{2\zeta(2\lambda)}{(2\pi^2)^{\lambda}} \right)^{|\mathbf{u}|} \right)^{1/\lambda} \quad \text{for all } \lambda \in (1/2, 1],$$

where $\zeta(x) := \sum_{k=1}^{\infty} k^{-x}$ denotes the Riemann zeta function for $x > 1$.

Remarks:

- Optimal rate of convergence $\mathcal{O}(n^{-1+\delta})$ in weighted Sobolev spaces, independently of s under an appropriate condition on the weights.
- Cost of algorithm for POD weights is $\mathcal{O}(s n \log n + s^2 n)$ using FFT.

Significance: Suppose that $f \in H_{s,\gamma}$ for all $\gamma = (\gamma_u)_{u \subseteq \{1:s\}}$. Then for any given sequence of weights γ , we can use the CBC algorithm to obtain a generating vector satisfying the error bound

$$\sqrt{\mathbb{E}_{\Delta} |I_s f - Q_{s,n}^{\Delta} f|^2} \leq \left(\frac{1}{\varphi(n)} \sum_{\emptyset \neq u \subseteq \{1:s\}} \gamma_u^{\lambda} \left(\frac{2\zeta(2\lambda)}{(2\pi^2)^{\lambda}} \right)^{|u|} \right)^{1/(2\lambda)} \|f\|_{s,\gamma} \quad (1)$$

for all $\lambda \in (1/2, 1]$. We can use the following strategy:

- For a given integrand f , estimate the norm $\|f\|_{s,\gamma}$.
- Find weights γ which *minimize* the error bound (1).
- Using the optimized weights γ as input, use the CBC algorithm to find a generating vector which *satisfies* the error bound (1).

Part II: Bayesian optimal experimental design

Let $G: \Theta \times \Xi \rightarrow \mathbb{R}^k$ be a forward mapping depending on a true parameter $\theta \in \Theta$ and a design parameter $\xi \in \Xi$.

Measurement model:

$$\mathbf{y} = G(\theta, \xi) + \eta,$$

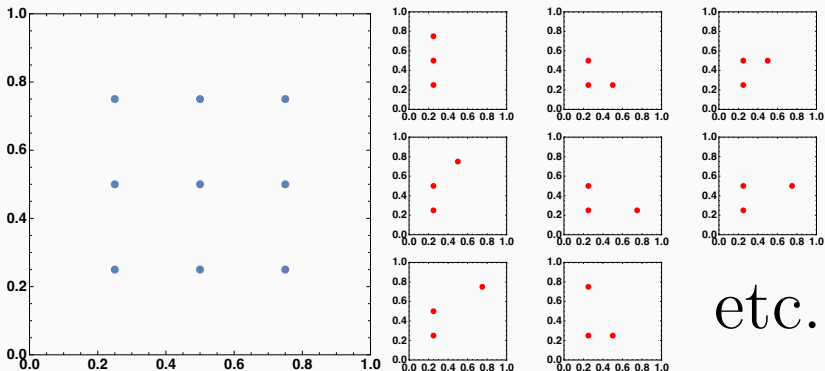
where $\mathbf{y} \in \mathbb{R}^k$ is the measurement data and $\eta \in \mathbb{R}^k$ is Gaussian noise such that $\eta \sim \mathcal{N}(0, \Gamma)$ with positive definite covariance matrix $\Gamma \in \mathbb{R}^{k \times k}$.

Goal in Bayesian optimal experimental design: Recover the design parameter ξ for the Bayesian inference of θ , which we model as a random variable endowed with prior distribution $\pi(\theta)$.

Example

Suppose we have 9 slots and 3 sensors. Before carrying out the experiment, which 3 slots do we expect to be the most informative for the recovery of the unknown parameter?

→ $\binom{9}{3} = 84$ possible configurations



Left: 9 slots. Right: We have 84 possible ways to place 3 sensors into 9 slots.

How to rank the 84 different possibilities from most informative to least informative?

A measure of the information gain for a given design ξ and data $\mathbf{y}$ is given by the Kullback–Leibler divergence

$$D_{\text{KL}}(\pi(\cdot|\mathbf{y}, \xi) \parallel \pi(\cdot)) := \int_{\Theta} \log \left(\frac{\pi(\boldsymbol{\theta}|\mathbf{y}, \xi)}{\pi(\boldsymbol{\theta})} \right) \pi(\boldsymbol{\theta}|\mathbf{y}, \xi) d\boldsymbol{\theta}. \quad (2)$$

We wish to maximize the expected utility (2) over the design space Ξ with respect to the data $\mathbf{y}$ and model parameters $\boldsymbol{\theta}$:

$$\max_{\xi \in \Xi} \underbrace{\int_{\mathcal{Y}} \int_{\Theta} \log \left(\frac{\pi(\boldsymbol{\theta}|\mathbf{y}, \xi)}{\pi(\boldsymbol{\theta})} \right) \pi(\boldsymbol{\theta}|\mathbf{y}, \xi) \pi(\mathbf{y}|\xi) d\boldsymbol{\theta} d\mathbf{y}}_{=:\text{EIG}},$$

where $\pi(\boldsymbol{\theta}|\mathbf{y}, \xi)$ corresponds to the posterior distribution of the parameter $\boldsymbol{\theta}$ and $\pi(\mathbf{y}|\xi) = \int_{\Theta} \pi(\mathbf{y}|\boldsymbol{\theta}, \xi) \pi(\boldsymbol{\theta}) d\boldsymbol{\theta}$ is the marginal distribution of the data $\mathbf{y}$.

The posterior is given by Bayes' theorem

$$\pi(\boldsymbol{\theta}|\mathbf{y}, \boldsymbol{\xi}) = \frac{\pi(\mathbf{y}|\boldsymbol{\theta}, \boldsymbol{\xi})\pi(\boldsymbol{\theta})}{\pi(\mathbf{y}|\boldsymbol{\xi})},$$

which means that the expected utility can be written as

$$\begin{aligned}\text{EIG} &= \int_{\mathbf{Y}} \int_{\boldsymbol{\Theta}} \log \left(\frac{\pi(\boldsymbol{\theta}|\mathbf{y}, \boldsymbol{\xi})}{\pi(\boldsymbol{\theta})} \right) \pi(\boldsymbol{\theta}|\mathbf{y}, \boldsymbol{\xi}) \, \mathrm{d}\boldsymbol{\theta} \, \pi(\mathbf{y}|\boldsymbol{\xi}) \, \mathrm{d}\mathbf{y} \\ &= \int_{\boldsymbol{\Theta}} \left[\int_{\mathbf{Y}} \log \left(\frac{\pi(\mathbf{y}|\boldsymbol{\theta}, \boldsymbol{\xi})}{\pi(\mathbf{y}|\boldsymbol{\xi})} \right) \pi(\mathbf{y}|\boldsymbol{\theta}, \boldsymbol{\xi}) \, \mathrm{d}\mathbf{y} \right] \pi(\boldsymbol{\theta}) \, \mathrm{d}\boldsymbol{\theta}.\end{aligned}$$

Approaches taken in the literature:

- Double-loop Monte Carlo (Beck, Mansour, Espath, Long, Tempone)
- MCLA (Beck, Mansour, Espath, Long, Tempone)
- DLMCIS (Beck, Mansour, Espath, Long, Tempone)

Let us assume the following:

A1 The forward model $\mathbf{y} = G(\boldsymbol{\theta}, \boldsymbol{\xi}) + \boldsymbol{\eta}$ satisfies

$$\|\partial_{\boldsymbol{\theta}}^{\boldsymbol{\nu}} G(\boldsymbol{\theta}, \boldsymbol{\xi})\| \leq C_0 |\boldsymbol{\nu}|! \mathbf{b}^{\boldsymbol{\nu}},$$

where $\mathbf{b} := (b_j)_{j \geq 1} \in \ell^p$ are nonnegative real numbers for some $p \in (0, 1)$ and $C_0 \geq 1$ is independent of $\boldsymbol{\xi} \in \Xi$.

A2 $\Theta = [-\frac{1}{2}, \frac{1}{2}]^s$ and $\pi(\boldsymbol{\theta}) = 1$ for $\boldsymbol{\theta} \in \Theta$ and 0 otherwise.

A3 The noise covariance is $\Gamma = \sigma^2 I_k$, $0 < \sigma \leq 1$.

Example

Model problem: Let $D \subset \mathbb{R}^d$, $d \in \{1, 2, 3\}$, be a nonempty, bounded, and convex Lipschitz domain and $z \in L^2(D)$. For each $\theta \in \Theta$, there exists a strong solution $u(\cdot, \theta) \in H^2(D) \cap H_0^1(D)$ to

$$\begin{cases} -\nabla \cdot (a(\mathbf{x}, \theta) \nabla u(\mathbf{x}, \theta)) = z(\mathbf{x}), & \mathbf{x} \in D, \theta \in \Theta, \\ u(\mathbf{x}, \theta) = 0, & \mathbf{x} \in \partial D, \theta \in \Theta, \end{cases}$$

where we assume that $\theta = (\theta_j)_{j \geq 1}$ i.i.d. uniformly distributed in $[-1/2, 1/2]$,

$$a(\mathbf{x}, \theta) = a_0(\mathbf{x}) + \sum_{j \geq 1} \theta_j \psi_j(\mathbf{x}), \quad \mathbf{x} \in D, \theta \in [-1/2, 1/2]^{\mathbb{N}},$$

with $a_0 \in W^{1,\infty}(D)$ and $\psi_j \in W^{1,\infty}(D)$, $j \geq 1$, such that

$\sum_{j \geq 1} \|\psi_j\|_{W^{1,\infty}(D)} < \infty$, $0 < a_{\min} \leq a(\mathbf{x}, \theta) \leq a_{\max} < \infty$ for all $\mathbf{x} \in D$, $\theta \in [-1/2, 1/2]^{\mathbb{N}}$, and $b_j := \|\psi_j\|_{L^\infty(D)} / a_{\min}$.

Then $G(\theta, \xi) := (u(\mathbf{x}, \theta))_{\mathbf{x} \in \xi}$, where $\xi = \{\mathbf{x}_1, \dots, \mathbf{x}_k\} \subset D$, satisfies A1.

If $\boldsymbol{\theta} \perp \boldsymbol{\eta}$, then the likelihood is given by

$$\pi(\mathbf{y}|\boldsymbol{\theta}, \boldsymbol{\xi}) = C e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - G(\boldsymbol{\theta}, \boldsymbol{\xi})\|^2}, \quad C = \frac{1}{(2\pi\sigma^2)^{k/2}}.$$

Under these conditions, it is easy to see that

$$\begin{aligned} \text{EIG} &= \int_{\mathbf{Y}} \int_{\boldsymbol{\Theta}} \log \left(\frac{\pi(\mathbf{y}|\boldsymbol{\theta}, \boldsymbol{\xi})}{\pi(\mathbf{y}|\boldsymbol{\xi})} \right) \pi(\mathbf{y}|\boldsymbol{\theta}, \boldsymbol{\xi}) \pi(\boldsymbol{\theta}) \, d\boldsymbol{\theta} \, d\mathbf{y} \\ &= \log C - 1 - \int_{\mathbf{Y}} \log \left(\int_{\boldsymbol{\Theta}} C e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - G(\boldsymbol{\theta}, \boldsymbol{\xi})\|^2} \, d\boldsymbol{\theta} \right) \int_{\boldsymbol{\Theta}} C e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - G(\boldsymbol{\theta}, \boldsymbol{\xi})\|^2} \, d\boldsymbol{\theta} \, d\mathbf{y}. \end{aligned}$$

Observations:

- The **inner integral** can be approximated *independently* of the dimension s using QMC.
- In general, the data dimension k affects the QMC cubature error bound of the **outer integral**.
- How to efficiently approximate the nested integrals?

Consider

$$\int_{\mathbf{y}} \log \left(\int_{\Theta} C e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - G(\boldsymbol{\theta}, \boldsymbol{\xi})\|^2} d\boldsymbol{\theta} \right) \int_{\Theta} C e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - G(\boldsymbol{\theta}, \boldsymbol{\xi})\|^2} d\boldsymbol{\theta} d\mathbf{y}.$$

- Parametric regularity of the integrand in

$$\int_{\Theta} C e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - G(\boldsymbol{\theta}, \boldsymbol{\xi})\|^2} d\boldsymbol{\theta}$$

is well-understood (as long as the parametric regularity of G can be quantified); straightforward modification of Herrmann–Keller–Schwab (2021).

- More generally, in many applications of UQ, we are interested in parametric integrals of quantities of the general form

$$f(G(\boldsymbol{\theta}, \cdot)), \quad (3)$$

where $f: \mathbb{R}^k \rightarrow \mathbb{R}$ is a (somewhat) smooth nonlinear quantity of interest[†], and

$$\|\partial_{\boldsymbol{\theta}}^{\boldsymbol{\nu}} G(\boldsymbol{\theta}, \cdot)\| \leq C_0 |\boldsymbol{\nu}|! \mathbf{b}^{\boldsymbol{\nu}}.$$

What can be said about the parametric regularity of (3)?

[†]E.g., $f(\mathbf{x}) = \|\mathbf{y} - \mathbf{x}\|^2$ or $f(\mathbf{x}) = e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{x}\|^2}$.

Faà di Bruno's formula:

$$\partial_{\theta}^{\nu} f(G(\theta, \cdot)) = \sum_{\substack{1 \leq |\lambda| \leq |\nu| \\ \lambda \in \mathbb{N}_0^k}} \partial_x^{\lambda} f(x) \Big|_{x=G(\theta, \cdot)} \kappa_{\nu, \lambda}(\theta), \quad \nu \neq \mathbf{0},$$

where the sequence $(\kappa_{\nu, \lambda})$ depends only on G via

$$\kappa_{\nu, \mathbf{0}} \equiv \delta_{\nu, \mathbf{0}},$$

$$\kappa_{\nu, \lambda} \equiv 0 \quad \text{if } |\nu| < |\lambda| \text{ or } \lambda \not\geq \mathbf{0} \text{ (i.e., if } \lambda \text{ contains negative entries),}$$

$$\kappa_{\nu + e_j, \lambda}(\theta) = \sum_{\ell \in \text{supp}(\lambda)} \sum_{\mathbf{0} \leq \mathbf{m} \leq \nu} \binom{\nu}{\mathbf{m}} \partial^{\mathbf{m} + e_j} [G(\theta, \cdot)]_{\ell} \kappa_{\nu - \mathbf{m}, \lambda - e_{\ell}}(\theta) \quad \text{otherwise.}$$

Since $\|\partial_{\theta}^{\nu} G(\theta, \cdot)\| \leq C_0 |\nu|! \mathbf{b}^{\nu}$, we obtain the following *uniform* bound.

Lemma

For all $1 \leq |\lambda| \leq |\nu|$, there holds

$$|\kappa_{\nu, \lambda}(\theta)| \leq C_0^{|\lambda|} \frac{|\nu|! (|\nu| - 1)!}{\lambda! (|\nu| - |\lambda|)! (|\lambda| - 1)!} \mathbf{b}^{\nu}.$$

Proof.

By induction w.r.t. the order of the multi-index ν .



For the present problem, plugging the upper bound for $|\kappa_{\nu,\lambda}(\theta)|$ into the expression

$$\partial_{\theta}^{\nu} f(G(\theta, \cdot)) = \sum_{\substack{1 \leq |\lambda| \leq |\nu| \\ \lambda \in \mathbb{N}_0^k}} \partial_x^{\lambda} f(x) \Big|_{x=G(\theta, \cdot)} \kappa_{\nu,\lambda}(\theta), \quad \nu \neq \mathbf{0},$$

and making some simple estimates yields altogether that

$$|\partial_{\theta}^{\nu} e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - G(\theta, \cdot)\|^2}| \leq 3.82^k \cdot C_0^{|\nu|} 2^{|\nu|-1} \sigma^{-|\nu|} |\nu|! \mathbf{b}^{\nu} \quad \text{for all } \nu \neq \mathbf{0}.$$

Decomposing the high-dimensional integral

Suppose that $Y = \mathbb{R}^k$. Then

$$\begin{aligned}
 & \int_{\mathbb{R}^k} \log \left(\int_{\Theta} C e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - G(\boldsymbol{\theta}, \boldsymbol{\xi})\|^2} d\boldsymbol{\theta} \right) \int_{\Theta} C e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - G(\tilde{\boldsymbol{\theta}}, \boldsymbol{\xi})\|^2} d\tilde{\boldsymbol{\theta}} d\mathbf{y} \\
 &= \int_{[-K, K]^k} \log \left(\int_{\Theta} C_{k, \sigma} e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - G(\boldsymbol{\theta}, \boldsymbol{\xi})\|^2} d\boldsymbol{\theta} \right) \int_{\Theta} C e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - G(\tilde{\boldsymbol{\theta}}, \boldsymbol{\xi})\|^2} d\tilde{\boldsymbol{\theta}} d\mathbf{y} \\
 &+ \int_{\mathbb{R}^k \setminus [-K, K]^k} \log \left(\int_{\Theta} C e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - G(\boldsymbol{\theta}, \boldsymbol{\xi})\|^2} d\boldsymbol{\theta} \right) \int_{\Theta} C e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - G(\tilde{\boldsymbol{\theta}}, \boldsymbol{\xi})\|^2} d\tilde{\boldsymbol{\theta}} d\mathbf{y} \\
 &=: \mathcal{I}_K + \tilde{\mathcal{I}}_K.
 \end{aligned}$$

For $K \gg 1$, there holds

$$\begin{aligned}
 |\tilde{\mathcal{I}}_K| &\leq C^{1/2} e^{\frac{\|\bar{G}\|^2}{4\sigma^2}} (4\pi\sigma^2)^{k/2} \\
 &- C^{1/2} e^{\frac{\|\bar{G}\|^2}{4\sigma^2}} (\pi\sigma^2)^{k/2} \prod_{j=1}^k \left(\operatorname{erf} \left(\frac{\bar{G}_j + K}{2\sigma} \right) - \operatorname{erf} \left(\frac{\bar{G}_j - K}{2\sigma} \right) \right),
 \end{aligned}$$

where $\bar{G} := (\bar{G}_j)_{j=1}^k := G(\boldsymbol{\theta}^*, \boldsymbol{\xi}^*)$, $(\boldsymbol{\theta}^*, \boldsymbol{\xi}^*) := \arg \max_{(\boldsymbol{\theta}, \boldsymbol{\xi}) \in (\Theta, \Xi)} \|G(\boldsymbol{\theta}, \boldsymbol{\xi})\|$.

$$\int_{[-K,K]^k} \log \left(\int_{\Theta} C e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - G(\boldsymbol{\theta}, \boldsymbol{\xi})\|^2} d\boldsymbol{\theta} \right) \int_{\Theta} C e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - G(\boldsymbol{\theta}, \boldsymbol{\xi})\|^2} d\boldsymbol{\theta} d\mathbf{y}$$

QMC weights for the **inner integral**:

$$\gamma_u = \left(|u|! \prod_{j \in u} \frac{\beta_j}{\sqrt{2\zeta(2\lambda)/(2\pi^2)^\lambda}} \right)^{\frac{2}{1+\lambda}}, \quad \lambda = \begin{cases} \frac{p}{2-p} & \text{if } p \in (2/3, 1), \\ \frac{1}{2-2\delta} & \text{if } p \in (0, 2/3], \end{cases}$$

with $\beta_j := \frac{2C_0}{\sigma} b_j$, $j \in \{1, \dots, s\}$, and $\delta > 0$ arbitrary, yields the QMC convergence rate

$$\mathcal{O}(\varphi(n)^{\max\{-1/p+1/2, -1+\delta\}})$$

independently of the dimension s with $\varphi(n)$ denoting the Euler totient function.

$$\int_{[-K,K]^k} \log \left(\int_{\Theta} C e^{-\frac{1}{2\sigma^2} \|\mathbf{y}-G(\boldsymbol{\theta},\boldsymbol{\xi})\|^2} d\boldsymbol{\theta} \right) \int_{\Theta} C e^{-\frac{1}{2\sigma^2} \|\mathbf{y}-G(\boldsymbol{\theta},\boldsymbol{\xi})\|^2} d\boldsymbol{\theta} d\mathbf{y}$$

QMC weights for the **outer integral**:

$$\tilde{\gamma}_u = \left(|u|! \prod_{j \in u} \frac{1.1^k k \sigma^{-1} e^{\frac{1}{2\sigma^2} (kK^2 + 2\sqrt{k}KC + C^2)}}{\log(2) \sqrt{2\zeta(2\tilde{\lambda})/(2\pi^2)^{\tilde{\lambda}}}} \right)^{\frac{2}{1+\tilde{\lambda}}}, \quad \tilde{\lambda} = \frac{1}{2 - 2\tilde{\delta}},$$

with $\tilde{\delta} > 0$ arbitrary, yields the QMC convergence rate

$$\mathcal{O}(\varphi(n)^{-1+\tilde{\delta}})$$

with $\varphi(n)$ denoting the Euler totient function. Note that the implied coefficient **depends on k** .

The nested integral

Goal of computation:

$$\mathcal{I}_K(f) = \int_{Y_K} g \left(\int_{\Theta} f(\boldsymbol{\theta}, \mathbf{y}, \boldsymbol{\xi}) d\boldsymbol{\theta} \right) d\mathbf{y},$$

where $g(x) := x \log x$, $Y_K = [-K, K]^k$, and $f(\boldsymbol{\theta}, \mathbf{y}, \boldsymbol{\xi}) := C e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - G(\boldsymbol{\theta}, \boldsymbol{\xi})\|^2}$.

Define a hierarchy of QMC cubature operators for the **outer integral**,
i.e.,

$$I^{(1)}F := \int_{Y_K} F(\mathbf{y}) d\mathbf{y} \approx 2^{-\ell} \sum_{k=1}^{2^\ell} F(\mathbf{y}_k^{(\ell)}) =: Q_\ell^{(1)}F, \ell = \ell_0^{(1)}, \ell_0^{(1)}+1, \ell_0^{(1)}+2, \dots,$$

for a given function $F \in \tilde{H}_{k, \tilde{\gamma}}$, and likewise for the **inner integral**

$$I^{(2)}F := \int_{\Theta} F(\boldsymbol{\theta}) d\boldsymbol{\theta} \approx 2^{-\ell} \sum_{k=1}^{2^\ell} F(\boldsymbol{\theta}_k^{(\ell)}) =: Q_\ell^{(2)}F, \ell = \ell_0^{(2)}, \ell_0^{(2)}+1, \ell_0^{(2)}+2, \dots,$$

for a given function $F \in H_{s, \gamma}$.

Why full tensor product cubature is a bad idea

Approximating the integral

$$\mathcal{I}_K(f) = \int_{Y_K} g \left(\int_{\Theta} f(\boldsymbol{\theta}, \mathbf{y}, \boldsymbol{\xi}) d\boldsymbol{\theta} \right) d\mathbf{y},$$

by

$$\mathcal{I}_K(f) \approx Q_{\ell}^{(1)} g(Q_{\ell}^{(2)} f) \tag{4}$$

is inefficient. A hand-wavy argument would be as follows:

- Suppose that we have the approximation rates (recall $n = 2^{\ell}$)

$$|I^{(1)}F - Q_{\ell}^{(1)}F| \asymp n^{-\alpha} \quad \text{and} \quad |I^{(2)}F - Q_{\ell}^{(2)}F| \asymp n^{-\alpha}.$$

- Evaluating (4) takes $N = n^2$ function calls, but the cubature accuracy will not be better than $\mathcal{O}(n^{-\alpha}) = \mathcal{O}(N^{-\alpha/2})$
→ the convergence rate is effectively halved! (“Curse of dimensionality”)

Sparse tensor product cubature in the vein of Gilch, Griebel, Oettershagen (2022)

Define the difference cubature operator corresponding to the **outer integral**

$$\Delta_{\ell}^{(1)} F := \begin{cases} Q_{\ell}^{(1)} F - Q_{\ell-1}^{(1)} F & \text{if } \ell \geq 1, \\ Q_0^{(1)} F & \text{if } \ell = 0, \end{cases}$$

as well as the *generalized* difference cubature operators corresponding to the **inner integral**

$$\Delta_{\ell}^{(2)} F := \begin{cases} g(Q_{\ell}^{(2)} F) - g(Q_{\ell-1}^{(2)} F) & \text{if } \ell \geq 1, \\ g(Q_0^{(2)} F) & \text{if } \ell = 0. \end{cases}$$

Generalized sparse grid cubature operator:

$$\mathcal{Q}_{L,\varsigma}(f) := \sum_{\varsigma \ell_1 + \frac{\ell_2}{\varsigma} \leq L} \Delta_{\ell_1}^{(1)} \Delta_{\ell_2}^{(2)}(f) = \sum_{\ell_1=0}^{L/\varsigma} \Delta_{\ell_1}^{(1)} g(Q_{\varsigma L - \varsigma^2 \ell_1}^{(2)} f).$$

Sparse tensor product cubature

$$\mathcal{Q}_{L,\varsigma}(f) := \sum_{\varsigma\ell_1 + \frac{\ell_2}{\varsigma} \leq L} \Delta_{\ell_1}^{(1)} \Delta_{\ell_2}^{(2)}(f) = \sum_{\ell_1=0}^{L/\varsigma} \Delta_{\ell_1}^{(1)} g(Q_{\varsigma L - \varsigma^2 \ell_1}^{(2)} f).$$

Sparse grid error: Our **inner** and **outer** QMC cubatures have essentially linear convergence rates, i.e.,

$$|I^{(1)}f - Q_\ell^{(1)}f| \lesssim 2^{-(1-\delta)\ell} \quad \text{and} \quad |I^{(2)}f - Q_\ell^{(2)}f| \lesssim 2^{-(1-\delta)\ell}.$$

For an *isotropic* ($\varsigma = 1$) sparse tensor product cubature operator, we obtain

$$\|\mathcal{I}_K(f) - \mathcal{Q}_{L,\varsigma}(f)\|_{\Delta} \lesssim 2^{-(1-\delta)L}(L+1)$$

under some additional technical assumptions.

Numerical experiment

Let $D = (0, 1)^2$. We consider the elliptic PDE

$$\begin{cases} -\nabla \cdot (a(\mathbf{x}, \boldsymbol{\theta}) \nabla u(\mathbf{x}, \boldsymbol{\theta})) = 10x_1, & \mathbf{x} \in D, \boldsymbol{\theta} \in [-1/2, 1/2]^{100}, \\ u(\cdot, \boldsymbol{\theta})|_{\partial D} = 0, & \boldsymbol{\theta} \in [-1/2, 1/2]^{100}, \end{cases}$$

equipped with the parametric diffusion coefficient

$$a(\mathbf{x}, \boldsymbol{\theta}) = 1 + 0.1 \sum_{j=1}^{100} j^{-2} \theta_j \sin(\pi j x_1) \sin(\pi j x_2), \boldsymbol{\theta} \in [-1/2, 1/2]^{100}.$$

Numerical experiment

The goal is to find a design ξ^* from the set

$$\Xi = \{(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) \in \Upsilon^3 \mid \mathbf{x}_i \neq \mathbf{x}_j \text{ for } i \neq j\},$$

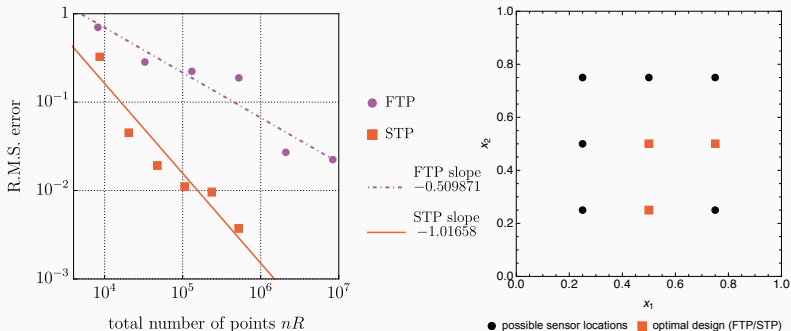
where

$$\begin{aligned} \Upsilon = \{ & (0.25, 0.25), (0.25, 0.50), (0.25, 0.75), \\ & (0.50, 0.25), (0.50, 0.50), (0.50, 0.75), \\ & (0.75, 0.25), (0.75, 0.50), (0.75, 0.75) \}, \end{aligned}$$

maximizing the expected information gain subject to the observation operator

$$G(\boldsymbol{\theta}, \boldsymbol{\xi}) = (u(\mathbf{x}, \boldsymbol{\theta}))_{\mathbf{x} \in \boldsymbol{\xi}}, \quad \boldsymbol{\theta} \in [-1/2, 1/2]^{100}, \quad \boldsymbol{\xi} \in \Xi.$$

Numerical experiment



Left: R.M.S. errors for the full tensor product (FTP) and sparse tensor product (STP) cubatures of the nested integral subject to affine and uniform parameterization of the input random field with $R = 8$ random shifts.

Right: the optimal design corresponding to the cubature rule with the largest number of points.

- QMC for DOE: sparse approach can recover almost the optimal rate.
- Preprint:
 -  V. Kaarnioja and C. Schillings. Quasi-Monte Carlo for Bayesian design of experiment problems governed by parametric PDEs. Preprint 2024. arXiv:2405.03529 [math.NA]

Thank you for your attention!